Final Project Report of Computational Methods and Modeling for Engineering Applications

GENG 8030 / Summer 2024

Adaptive Cruise Control

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ABSTRACT

Adaptive Cruise Control (ACC) is an innovative automotive technology designed to significantly improve driving safety and comfort. Utilizing advanced sensors, cameras, radar systems, and real-time data analysis, ACC maintains safe distances between vehicles, thereby reducing the risk of accidents and enhancing traffic flow. The system provides drivers with the flexibility to activate or deactivate it according to their preferences, ensuring a high degree of control over the driving experience. As automotive manufacturers continue to enhance ACC's capabilities, it is set to become a vital component of smart transportation systems, promising a future of safer and more efficient driving.

1. INTRODUCTION

Adaptive Cruise Control (ACC) is a cutting-edge technology developed to ensure a safe distance between your vehicle and the one ahead. Part of the advanced driver assistance systems (ADAS), ACC utilizes an array of sensors, cameras, and radars to constantly assess the surrounding environment and adjust the vehicle's speed as necessary. Its primary purpose is to enhance driving comfort and safety by responding to varying traffic conditions.

By alleviating driver stress, enhancing driving efficiency, and mitigating accidents due to human errors like tailgating or not adapting to changing traffic speeds, ACC is crucial for modern vehicle safety. It maintains a safe gap between vehicles, thereby preventing rear-end collisions and lessening the impact of potential accidents. As an active safety feature, ACC steps in when needed to avoid or minimize collisions, representing a major advancement in automotive technology.

Figure 1. Adaptive Cruise Control Model [1]

Adaptive Cruise Control (ACC) plays a pivotal role in enhancing road network efficiency. By minimizing abrupt braking and acceleration, it contributes to smoother traffic flow and reduces congestion. The ultimate goal of ACC is to improve driving safety, comfort, and efficiency for all road users, including both drivers and passengers.

Modern ACC systems are advanced enough to automatically adjust speeds when navigating sharp curves, significantly reducing the risk of accidents. Drivers have the option to switch over to activate mode or deactivate mode as needed, giving them more control over their driving experience.

As road safety becomes an increasing priority and the development of autonomous vehicles progresses, ACC technology is rapidly advancing. Manufacturers are heavily investing in the advanced development of ACC systems that can operate in any environment, from city streets to rural roads as the importance of ACC is in demand. Figure[1] provides a pictorial representation of ACC.

2. HARDWARE DESCRIPTION: COMPONENTS

2.1 Arduino UNO

The ATmega328P is a very adaptable microcontroller that powers the Arduino Uno, one of the most well-known development boards in the Arduino ecosystem. This 8-bit AVR RISC-based microcontroller has 32 KB of flash memory, 2 KB of SRAM, and 1 KB of EEPROM. With a clock speed of 16 MHz, the ATmega328P delivers impressive performance for its size. The Arduino Uno uses this microcontroller to provide 14 digital I/O pins (6 of which allow PWM output), 6 analogue input pins, and a variety of interfaces such as UART, SPI, and I2C. The board operates at 5V and is noted for its simplicity, making it a favourite among professionals alike for prototyping and experimenting with electronic projects.

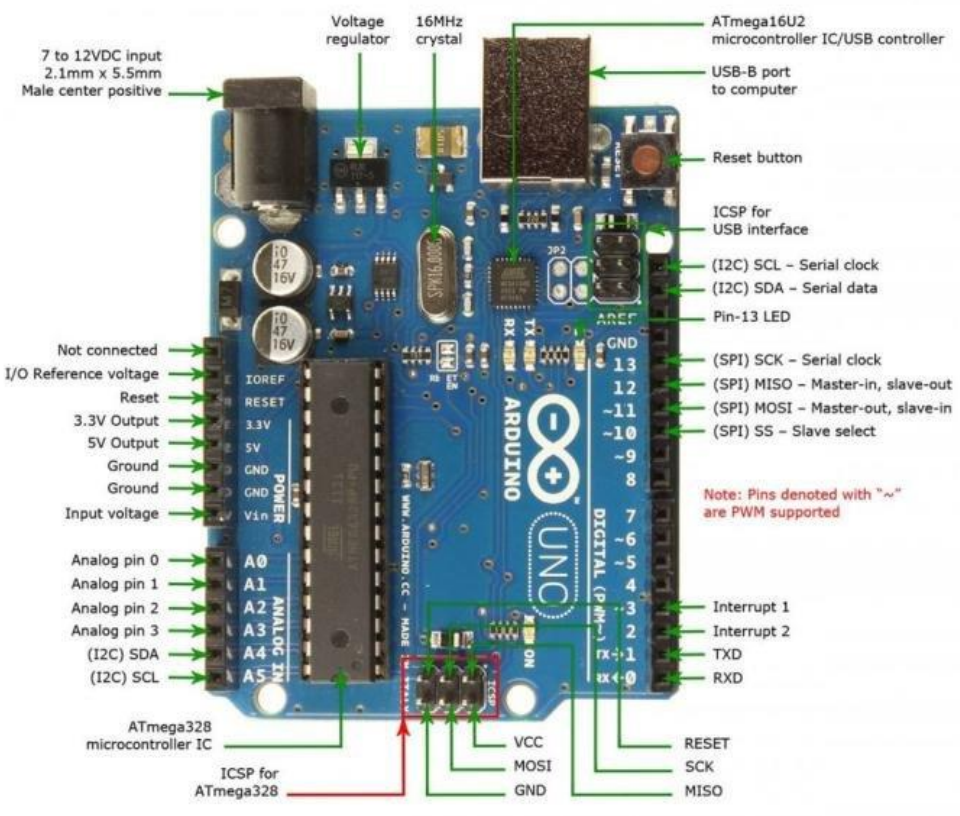

Figure 2. Arduino UNO [2]

SPECIFICATIONS

● Microcontroller ATmega328

● Operating Voltage 5V

● Input Voltage (recommended) 7-12V

● Input Voltage (limits) 6-20V

● Digital I/O Pins 14 (of which 6 provide PWM output)

● Analog Input Pins 6

● DC Current per I/O Pin 40 mA

● DC Current for 3.3V Pin 50 mA

● Flash Memory 32 KB (ATmega328) of 0.5 KB

● SRAM 2 KB (ATmega328)

● EEPROM 1 KB (ATmega328)

● Clock Speed 16 MHZ

2.2 Ultrasonic Sensor

The ultrasonic sensor is a device with electronics that determines the distance of a target object by emitting ultrasonic sound waves and converting what it receives into electrical signals. These sensors, such as sonar and radar, evaluate sound wave reflections to determine a target's attributes. The sensor consists of two main components: a transmitter, which emits sound with piezoelectric crystals, and the other componenet called a receiver, which detects the waves that bounce back from the object. The distance to the target is determined using the period that is required for the signal to go from the device that transmits to the device that receives it. They are commonly employed in a range of sectors and fields because of their capacity to measure and identify things or distances without physical contact. [3].

Figure 3. Ultrasonic Sensor [3]

2.3 Resistors

Resistors are basic electronic components that regulate the passage of electrical current. The resistance in ohms (Ω) indicates how strongly they reject current. They are responsible for

regulating current, designing voltage dividers, determining operating points for active devices, and maintaining correct logic levels in digital circuits. Resistors tolerance, power rating, and resistance value are all key factors that influence their performance and appropriateness for a wide range of applications.

Figure 4. Resistor [3]

2.4 Breadboard

A breadboard is a tool for developing semi-permanent electrical circuit prototypes. Breadboards do not require soldering or track destruction; therefore, they are reusable. This capacity has made breadboards popular in classrooms and among students who are learning technology. Breadboards have a grid of holes through which circuit components like as resistors, capacitors, and integrated circuits can be inserted1. These holes are perfectly aligned internally, allowing components to be simply attached by plugging their leads into the appropriate holes [5].

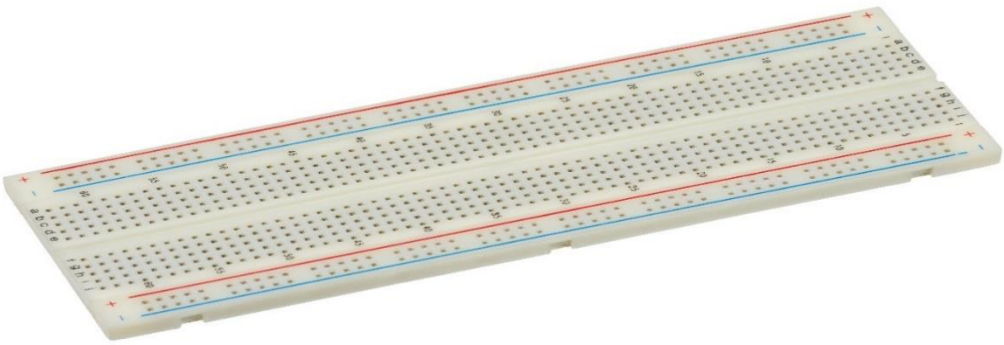

Figure 5. Breadboard [5]

2.5 LCD (Liquid Crystal Display)

LCDs (Liquid Crystal Displays) are key components of adaptive cruise control systems, providing the driver with real-time information and visual feedback. LCDs are flat panel displays that create images by modulating lights employing the use of liquid crystals. These crystals don't emit light directly. Instead, they use a backlight or reflector to show images in colour or monochrome. These displays offer vital information such as the set speed, following distance, and cruise control system status, keeping the driver informed of the vehicle's performance.

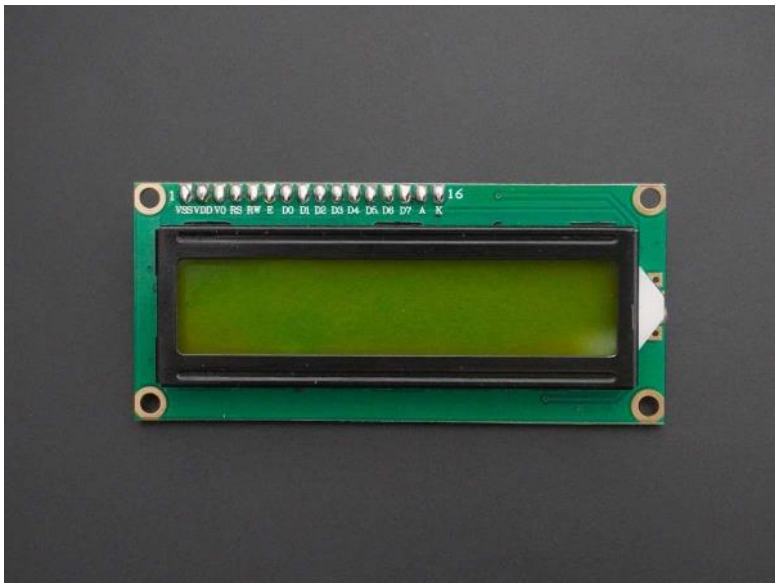

Figure 6. Liquid Crystal Display (LCD) [6]

2.6 Button

The button component in an adaptive cruise control system is essential for human interaction, acting as a tactile interface for operating the system's many functions. Each button is usually a momentary switch, which means it only has one state when pressed. The system has five buttons. These controls enable the driver to choose a cruising pace, activate adaptive cruise control mode, exit the current mode, and adjust the speed up or down.

Figure 7. Button [7]

2.7 Jumper Wires

Jumper wires are required for designing and building electronic circuits, especially when utilizing breadboards and microcontrollers such as the Arduino. These flexible wires allow you to connect multiple components in a circuit without the need for solder. Jumper wires are essential throughout the development and prototyping stages of electronic projects because they provide a quick and dependable method of connecting components, allowing for easy circuit testing, modification, and troubleshooting.

Figure 8. Jumper Wires [8]

3. SYSTEM SOFTWARE DESCRIPTION: MATLAB

MATLAB (Matrix Laboratory) is a sophisticated programming environment known for its high-level interactive platform, widely employed for numerical computation, data visualization, and programming. Created by MathWorks, MATLAB is particularly favored in engineering, scientific research, and applied mathematics due to its robust computational power

and versatile toolsets. Below is a summary of MATLAB's key features and functionalities relevant to system software development:

Interactive Environment:

MATLAB offers an interactive workspace with a command window, workspace management, and advanced debugging tools. Users can execute commands, run scripts, and visualize data in real-time, enabling rapid prototyping and iterative improvements.

Comprehensive Mathematical Functions:

The software includes a broad library of built-in functions for various mathematical operations, such as linear algebra, statistical analysis, Fourier transforms, filtering, optimization, numerical integration, and solving ordinary differential equations (ODEs). These functions facilitate efficient and precise mathematical modeling and analysis.

Visualization Capabilities:

MATLAB is highly regarded for its data visualization capabilities, providing tools for creating 2D and 3D plots, graphs, and charts. This functionality enhances the understanding and presentation of complex datasets.

Simulink Integration:

Simulink, an add-on to MATLAB, offers a graphical environment for modeling, simulating, and analyzing dynamic systems. It is extensively used for designing control systems, signal processing, and developing embedded systems.

Specialized Toolboxes:

MATLAB includes various specialized toolboxes tailored for different applications, such as signal processing, image processing, control systems, and machine learning. These toolboxes extend MATLAB's core functionality to specific domains.

Programming and Scripting:

MATLAB supports programming with its own high-level language, featuring loops, conditional statements, functions, and data structures. Users can write scripts and functions to automate tasks, conduct complex calculations, and develop custom algorithms.

Interoperability:

MATLAB can interface with other programming languages like C, C++, Java, Python, and Fortran, allowing users to integrate existing code and libraries. It also supports various data import and export formats, facilitating integration with external applications and systems.

Code Generation:

MATLAB Coder allows automatic generation of C and C++ code from MATLAB algorithms, enabling deployment to embedded systems and high-performance computing environments.

App Development:

MATLAB provides tools for developing custom graphical user interfaces (GUIs) using App Designer, allowing the creation of interactive applications for data analysis, visualization, and control.

Application in System Development:

For the development of an Adaptive Cruise Control (ACC) system, MATLAB offers significant advantages:

-Modeling and Simulation: Engineers can model vehicle dynamics, sensor behavior, and control algorithms within MATLAB and Simulink, facilitating thorough testing and validation of the ACC system.

Algorithm Development: The extensive mathematical functions and toolboxes in MATLAB support the development of complex control algorithms required for adaptive cruise control.

Data Analysis: MATLAB's data analysis and visualization tools are essential for examining sensor data, fine-tuning control parameters, and optimizing system performance.

-Prototyping and Testing: The interactive environment of MATLAB allows for rapid prototyping and iterative testing of control strategies, ensuring reliable system performance before real-world implementation.

In summary, MATLAB provides a powerful and flexible platform for developing, testing, and deploying advanced system software, making it an essential tool in the development of an Adaptive Cruise Control system.

4. PROJECT DESCRIPTION

4.1 PROBLEM STATEMENT:

The development of an Adaptive Cruise Control (ACC) system aims to enhance vehicle safety and driving convenience by maintaining a safe distance from other vehicles and adhering to speed limits. The current system integrates several functionalities, including speed control features, adaptive cruise control, and a user-friendly interface. However, challenges persist in optimizing the seamless transition between regular and adaptive cruise control modes, ensuring reliable sensor detection and response, and maintaining user control and understanding of the system's functionalities. This project seeks to address these issues by improving the automation features and providing intuitive controls to enhance the overall driving experience and safety.

4.2 PROBLEM SOLUTION:

The Adaptive Cruise Control (ACC) system development focuses on enhancing vehicle safety by maintaining a safe distance from other vehicles while adhering to speed limits. This system integrates several key functionalities to optimize driving comfort and safety:

1. Speed Control Features:

- Users can increase the vehicle speed by pressing the speed-up button.
- Decreasing the vehicle speed is easily achieved using the speed down button.
- Cruise control functionality allows for setting and maintaining a constant vehicle speed, with the ability to exit this mode via the cancel button.

2. Adaptive Cruise Control (ACC):

- Activating adaptive cruise control involves pressing its dedicated button. Once activated, the system uses distance sensors to monitor the vehicle ahead and automatically adjusts the vehicle's speed accordingly.
- When an object is detected within a predefined distance threshold, the ACC system initiates automatic speed reduction. Upon clearing the path ahead, it smoothly accelerates back to the desired speed.
- Adaptive mode operates independently of the speed control buttons used in regular cruise control mode.

3. User Interface and Functionality:

- Upon activation, the system initializes with a default speed display of 0. Users can adjust this initial speed using the provided speed control buttons.

- Cruise control mode can be engaged by pressing the set button, enabling the vehicle to maintain a steady speed until manually adjusted or cancelled.
- Clear transitions between adaptive cruise control and regular cruise control modes are indicated on the display through flashing notifications.

This project aims to enhance driving convenience and safety through advanced automation features while providing intuitive controls for users to manage and optimize their driving experience.

5. BLOCK DIAGRAM

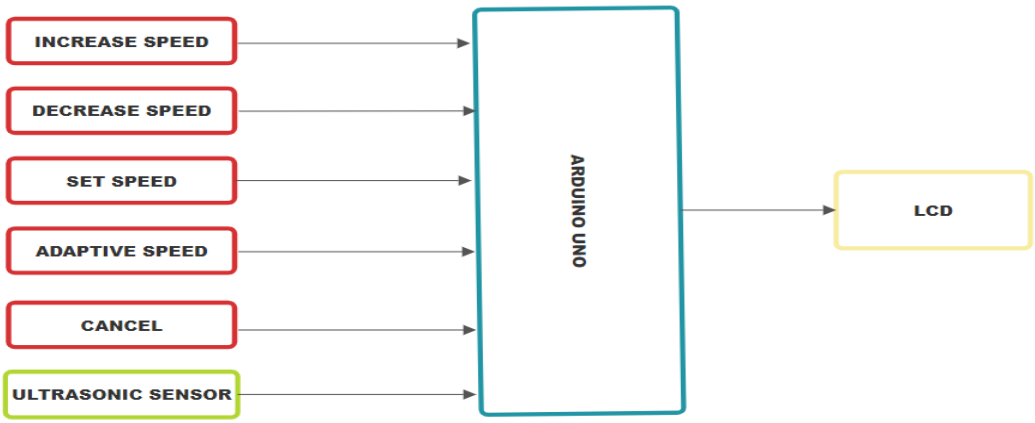

Figure 9. Block Diagram

6.CIRCUIT DIAGRAM:

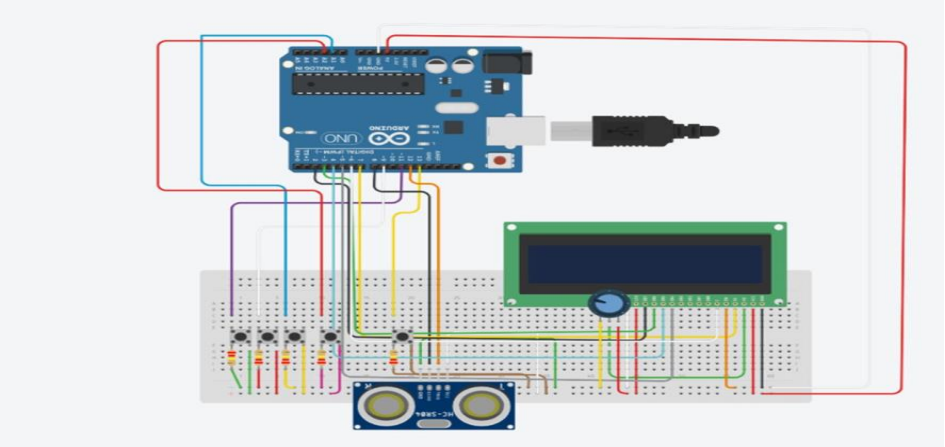

Figure 10. Circuit diagram

7.FLOWCHART

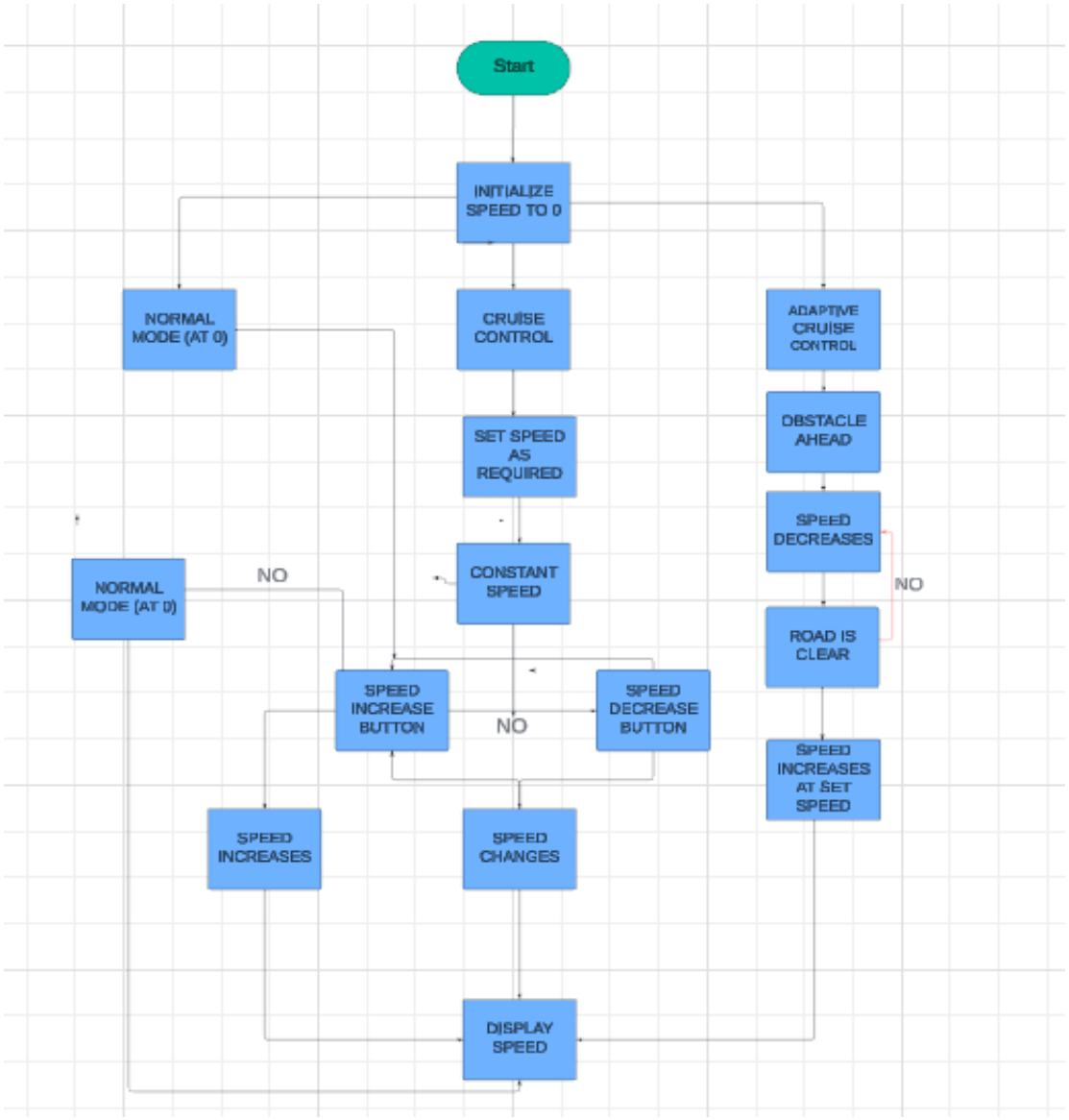

Figure 11. Flowchart

8.CIRCUIT CONNECTIONS:

8.1 LCD AND ARDUINO CONNECTIONS

LCD Pin	Arduino Pin	Purpose
VSS	GND	Ground Pin for connection
VDD	+5V	Power Source
V0	Potentiometer Input	Control pin to adjust contrast of LCD display
RS	8	Toggle between Command and Data mode
RW	GND	Toggle between Read and Write mode
E	13	Must be held high to perform Read/Write Operation
D0	NA	NA
D1	NA	NA
D2	NA	NA
D3	NA	NA
D4	12	Data bit pins to send command or data to LCD
D5	11	
D6	10	
D7	9	
A	+5V	Normal LED like operation to illuminate the LCD
K	GND	Normal LED-like operation to illuminate the LCD connected with GND

Table 1 LCD AND ARDUNO CONNECTIONS

8.2 Ultrasonic Sensor to Arduino Connections

Ultrasonic Sensor Pin	Arduino Pin	Purpose
GND	GND	Ground pin for connection
ECHO	5	This output pin becomes active for a set duration, corresponding to the time it takes for the wave to travel back to the sensor
TRIG	4	This input pin is used to send ultrasonic waves and initialise measurements by maintaining it high for 10us
VCC	+5V	Power sourcefPS

Table 2 Ultrasonic Sensor connections

9.MATLAB CODE:

```
clear;

clc;


% Define pins

Bon_ine_sed = 'D6'; % Increase speed button

Bon_dee_sed = 'D7'; % Decrease speed button

Sed_bon = 'D8';     % Set speed button

bon_ccl = 'D9';     % Cancel button

Bon_adape = 'D10';  % Adaptive speed button

P_TRIG = 'A0';

PIN_ECHO = 'A1';

rs = 'D12'; en = 'D11'; d4 = 'D5'; d5 = 'D4'; d6 = 'D3'; d7 = 'D2';


% Initialize Arduino and add-ons

ard_obj = arduino('COM5', 'Uno', 'Libraries', {'ExampleLCD/LCDAddon', 'Ultrasonic'},
'ForceBuildOn', true);

lcd = addon(ard_obj, 'ExampleLCD/LCDAddon', 'RegisterSelectPin', rs, 'EnablePin', en,
'DataPins', {d4, d5, d6, d7});

uls_Obj = ultrasonic(ard_obj, P_TRIG, PIN_ECHO, "OutputFormat", "double");



initializeLCD(lcd);

clearLCD(lcd);

printLCD(lcd, 'Welcome !');

pause(2.5);

clearLCD(lcd);
```

```
printLCD(lcd, 'ACC');

pause(2.5);

clearLCD(lcd);

printLCD(lcd, 'Suruthi');

pause(2);

clearLCD(lcd);

printLCD(lcd, '110150851');

pause(2);

clearLCD(lcd);

printLCD(lcd, 'Santhosh');

pause(2);

clearLCD(lcd);

printLCD(lcd, '110153849');

pause(2);

clearLCD(lcd);

printLCD(lcd, 'Ajith');

pause(2);

clearLCD(lcd);

printLCD(lcd, '110153123');

pause(2);

clearLCD(lcd);


Inre = false;

dece = false;

Spst = false;

Adset = false;
```

```
Spdtemp = 0;
VEHI_SPD = 0;
timer_stop = 0.2;

% Setup
configurePin(ard_obj, Bon_ine_sed, 'Pullup');
configurePin(ard_obj, Bon_dee_sed, 'Pullup');
configurePin(ard_obj, Sed_bon, 'Pullup');
configurePin(ard_obj, bon_ccl, 'Pullup');
configurePin(ard_obj, Bon_adape, 'Pullup');

initializeLCD(lcd);
clearLCD(lcd);
printLCD(lcd, strcat('VEHI_SPD: ', num2str(VEHI_SPD)));

% Loop
while true
  dis_tance = 0;

  % Enter adaptive speed mode
  if (readDigitalPin(ard_obj, Bon_adape) == 0)
    Spdtemp = VEHI_SPD;
    Adset = true;
    Spst = true;
    Inre = false;
```

```
    dece = false;
end

% Adaptive speed mode logic
if Adset
    dis_tance = readDistance(uls_Obj) * 100;
    disp(dis_tance)
    disp_adapt(lcd, VEHI_SPD);
end

% Adjust speed based on distance in adaptive mode
if dis_tance < 20 && Adset
    while dis_tance < 20 && readDigitalPin(ard_obj, bon_ccl) ~= 0
        if VEHI_SPD > 0
            VEHI_SPD = VEHI_SPD - 1;
        end
        disp_adapt(lcd, VEHI_SPD);
        dis_tance = readDistance(uls_Obj) * 100;
    end
    while dis_tance > 20 && VEHI_SPD < Spdtemp && readDigitalPin(ard_obj, bon_ccl) ~= 0
        VEHI_SPD = VEHI_SPD + 1;
        disp_adapt(lcd, VEHI_SPD);
        dis_tance = readDistance(uls_Obj) * 100;
        pause(0.3);
    end
    pause(0.2);
```

```
end

% Increase speed button logic outside adaptive mode
if readDigitalPin(ard_obj, Bon_ine_sed) == 0 && ~Adset
  timer_stop = 0.2;
  while readDigitalPin(ard_obj, Bon_ine_sed) == 0 && ~Spst
    VEHI_SPD = VEHI_SPD + 1;
    disp_updt(lcd, VEHI_SPD);
    pause(timer_stop);
  end
  if ~Spst
    Inre = false;
    dece = true;
    timer_stop = 1;
    pause(timer_stop);
  else
    dece = false;
    Inre = true;
    timer_stop = 0.2;
    pause(timer_stop);
  end
end

% Decrease speed button logic outside adaptive mode
if readDigitalPin(ard_obj, Bon_dee_sed) == 0 && ~Adset
```

```
    timer_stop = 0.2;
    Inre = false;
    dece = true;
    pause(timer_stop);
end

% Set speed button logic
if readDigitalPin(ard_obj, Sed_bon) == 0
    timer_stop = 0.2;
    Spst = true;
    Inre = false;
    dece = false;
    pause(timer_stop);
end

% Cancel button logic
if readDigitalPin(ard_obj, bon_ccl) == 0
    timer_stop = 1;
    Spst = false;
    Adset = false;
    Inre = false;
    dece = true;
    pause(timer_stop);
end

% Increase speed while not in set speed mode
```

```
if Inre && ~Spst
   VEHI_SPD = VEHI_SPD + 1;
   disp_updt(lcd, VEHI_SPD);
   pause(timer_stop);
end

% Decrease speed while not in set speed mode
if dece && ~Spst
   if VEHI_SPD > 0
      VEHI_SPD = VEHI_SPD - 1;
   else
      dece = false;
   end
   disp_updt(lcd, VEHI_SPD);
   pause(timer_stop);
end

% Increase speed while in set speed mode
if Inre && Spst
   VEHI_SPD = VEHI_SPD + 1;
   Inre = false;
   disp_updt(lcd, VEHI_SPD);
   pause(timer_stop);
end
```

```
  % Decrease speed while in set speed mode
  if dece && Spst
    if VEHI_SPD > 0
      VEHI_SPD = VEHI_SPD - 1;
    end
    dece = false;
    disp_updt(lcd, VEHI_SPD);
    pause(timer_stop);
  end

  % Slow speed change if no set speed button pressed and not in adaptive mode
  if ~Spst && ~Adset
    if VEHI_SPD > 0
      VEHI_SPD = VEHI_SPD - 0.1;
    end
    disp_updt(lcd, VEHI_SPD);
    pause(0.5);
  end
end

% Function to update display in normal mode
function disp_updt(lcd, VEHI_SPD)
  clearLCD(lcd);
  printLCD(lcd, strcat('VEHI_SPD: ', num2str(VEHI_SPD)));
end
```

```
% Function to update display in adaptive mode with blinking effect

function disp_adapt(lcd, VEHI_SPD)

   clearLCD(lcd);

   pause(0.5);

   printLCD(lcd, strcat('VEHI_SPD: ', num2str(VEHI_SPD)));

   pause(0.5);

end
```

10. PICTURE OF SETUP AND WIRING

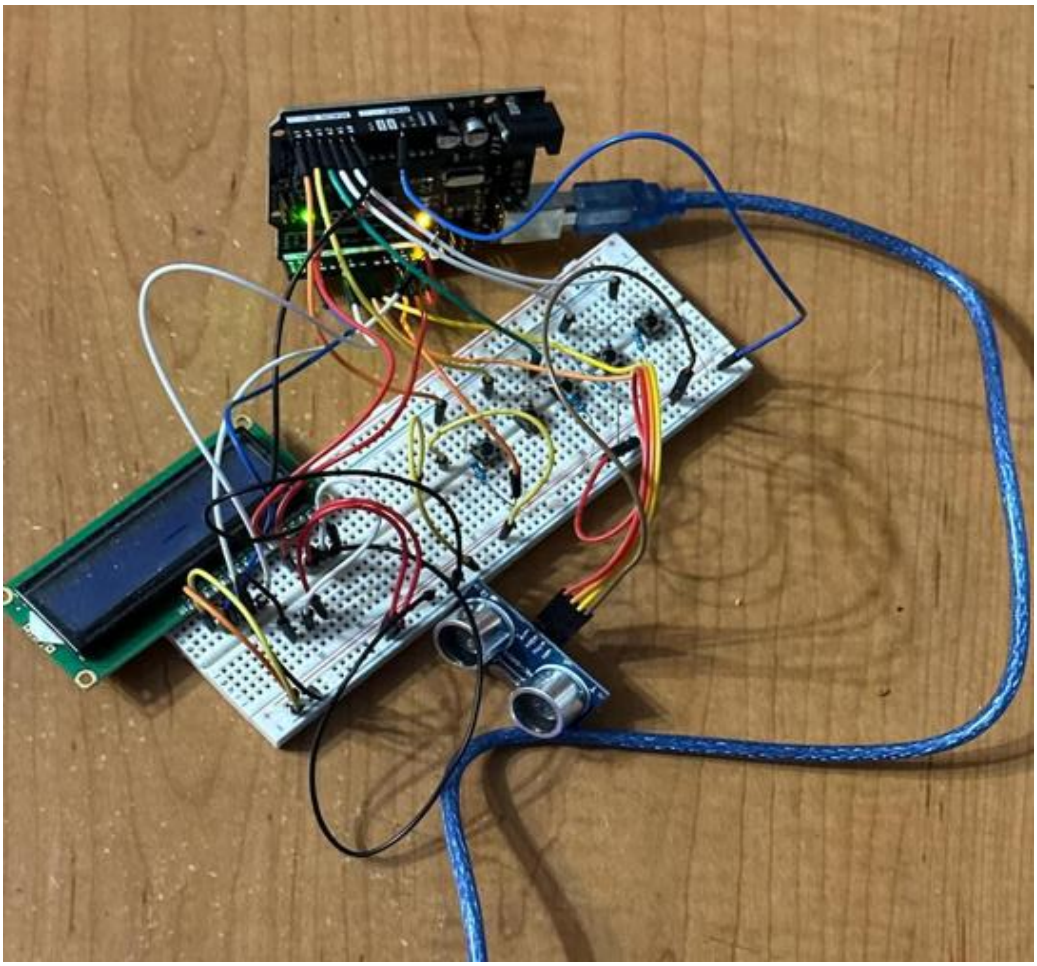

Figure 12. SETUP AND WIRING

11. CONCLUSION:

In conclusion, the goal of this project is to improve driving economy, comfort, and safety, which is a major advancement in automobile technology. ACC systems reduce the likelihood of accidents and facilitate smoother traffic flow by maintaining safe distances between cars via the use of sophisticated sensors, cameras, and real-time data processing. The system's ability to be turned on or off at will by the driver gives them complete control over their driving experience, accommodating different driving situations and preferences.

The addition of ACC to contemporary cars demonstrates how intelligent transportation systems are continually developing and demonstrates a dedication to safer and more effective roads. We may predict even greater breakthroughs in vehicle automation, which will contribute to the future of autonomous driving, as automakers continue to invent and improve ACC technology. In general, ACC is certain to become a standard feature of cars of the future, improving both road safety and driving comfort.

12. REFERENCE:

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